Research on Currency Volatility Forecasting Based on Machine Learning Models

— A Case Study of Cryptocurrencies and Traditional Currencies

Reporting Group: Group 8

Contents

Research Background

Brief introduction of research background, data sources, and hypotheses.

01 Research Background：Foreword

Core of Risk Management

Exchange rate and cryptocurrency price volatility have a direct impact on financial stability, and accurate volatility forecasting is crucial for investment decisions and risk management.

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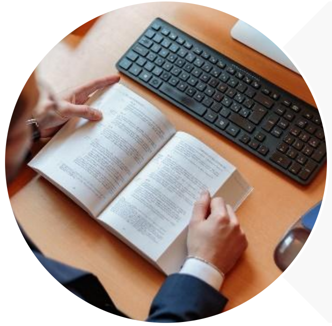

Market uncertainty

The uncertainty and instability in global financial markets have significantly intensified, with currency volatility risks remaining prominent. Conducting volatility forecasting research holds substantial practical significance.

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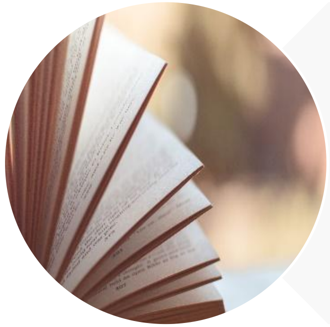

Update prediction method

Machine learning demonstrates remarkable advantages in nonlinear time series forecasting, offering innovative approaches for modeling currency volatility, with substantial research potential remaining to be explored.

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Traditional currency：

AUDUSD/EURUSD/GBPUSD

Cryptocurrency：

BTCUSD/ETHUSD

Data range：

The dataset covers hourly and daily data from December 12,2017 to March 9, 2026, with ample samples to capture fluctuation patterns across various market conditions.

Hypothesis：

Model fusion validity assumption： Hybrid models (such as GARCH-XGBoost) demonstrate significantly superior prediction accuracy for volatility compared to standalone machine learning models.

Hypothesis on the Impact of Asset Heterogeneity： The hybrid model demonstrates varying performance in volatility prediction for two asset classes, with differing improvement magnitudes observed.

Dynamic Correlation and Time-Varying Assumption： The volatility correlation between cryptocurrencies and traditional currencies is not constant but dynamically evolving (time-varying). Furthermore, major external shocks (such as the COVID-19 pandemic) can significantly alter the magnitude and structure of this correlation, leading to enhanced cross-market risk linkages.

Feature Comparison

By drawing time series graphs of **closing prices**, **historical volatility** and **logarithmic returns**, the characteristics of volatility of cryptocurrencies and traditional currencies can be visually compared.

The closing prices, logarithmic returns, and historical volatility trends of Bitcoin and Ether are **nearly identical**, and there is a **high degree of correlation** between the two.

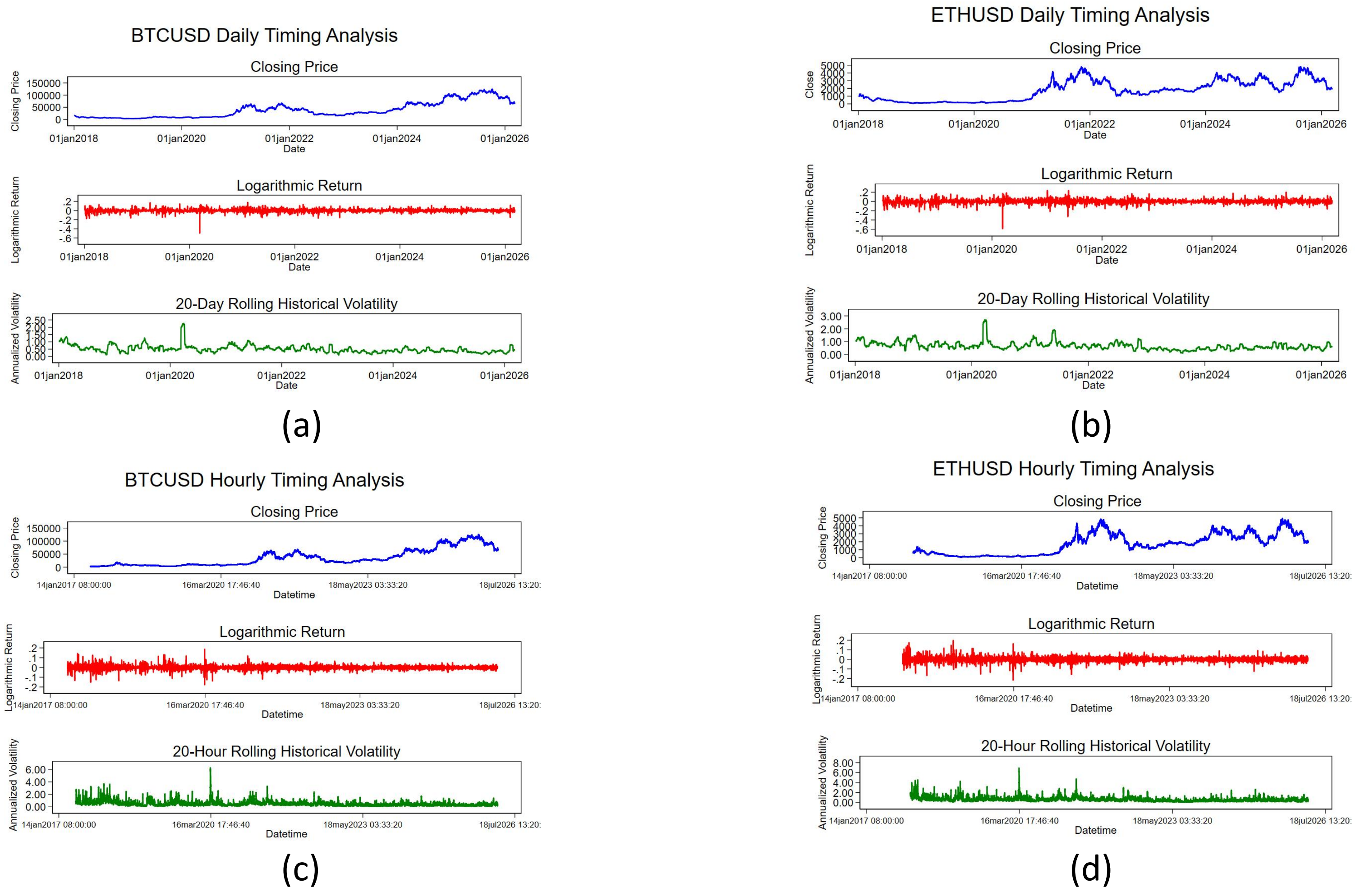

Figure 1： BTCUSD/ETHUSD Daily and Hourly Analysis

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Table 1: Two Typical Effects

	leverage effect	clustering effect
Phenomenon	• **Cryptocurrency exhibits more pronounced leverage effects**: When prices decline, volatility rises significantly more than traditional currencies, resulting in greater negative impacts. • **High-frequency data effect weakening**: In hourly high-frequency data, the leverage effect of both asset classes has shown a decline.	• **Cryptocurrencies exhibit stronger agglomeration characteristics:**The volatility exhibits pronounced clustering characteristics, with high and low volatility events clustering together, and extreme tail risks significantly exceeding those of traditional currencies. • **The high-frequency data effect is more pronounced:**At the hourly time scale, volatility clustering becomes more pronounced, with high-volatility events occurring frequently but exhibiting shorter durations.
Explanation	• **Cryptocurrencies: Speculation and Panic** The market exhibits strong speculative characteristics with investors demonstrating high risk appetite. Panic selling during market declines exacerbates volatility. • **Traditional currencies: Regulation and maturity** Subject to central bank monetary policy adjustments, these currencies exhibit mature market mechanisms, strong ability to absorb negative news, and stable volatility.	• **Cryptocurrencies: Sentiment-driven dynamics and lack of intervention** Market sentiment tends to spread and amplify rapidly, leading to "buying high and selling low" behavior; without official intervention, price volatility is unlikely to revert to the mean. • **Traditional currencies: Policy intervention and market depth** Monetary policy interventions by central banks effectively mitigate volatility; foreign exchange markets exhibit significant depth and breadth, possessing self-regulating capabilities.
Conclusion	The differences in leverage effects fundamentally reflect **distinct market structures** of two asset classes and **divergent investor behaviors**.	The probability of concentrated outbreak of **cryptocurrency market risks in the short term is significantly higher** than that of traditional money markets.

02 Feature comparison: External shock

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The pandemic is a **long-term trend-driven** event for **cryptocurrencies**. After a short-term extreme shock, **loose liquidity and market demand** resonate to create a super bull market, and the long-term central points of price and volatility both move upward. The hourly data can better reflect its **"high-frequency high volatility + strong trend"** characteristics.

The pandemic is a **short-term exogenous shock** for **traditional currencies**, which does not change their fundamental driving logic. **Prices and volatility both quickly return to the long-term equilibrium**, and the impact responses of daily and hourly data are consistent (only the hourly data has more detailed fluctuations).

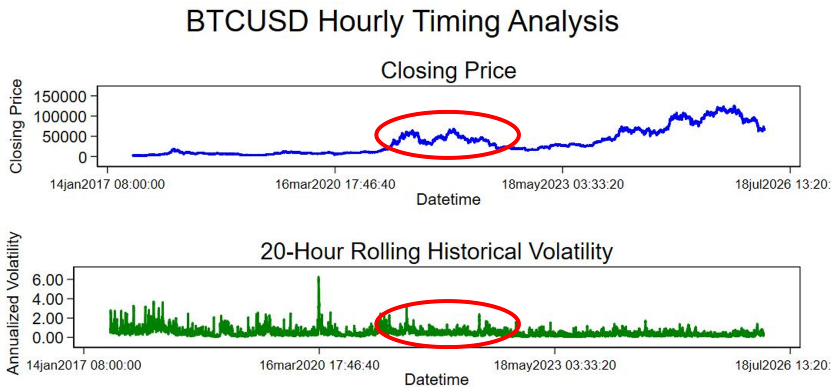

Figure 2： BTCUSD Hourly Analysis

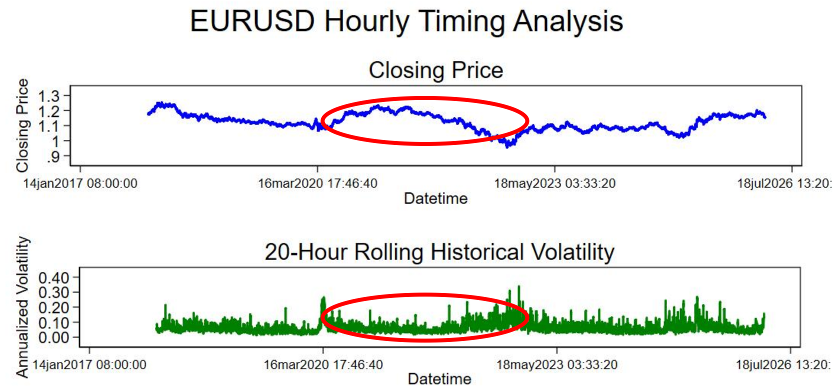

Figure 3： EURUSD Hourly Analysis

Conclusion: For traditional currencies, the pandemic was merely a short-term disruption, with markets quickly returning to normal. However, for cryptocurrencies, **the pandemic became a pivotal turning point.** It not only triggered significant short-term volatility but, more importantly, reshaped their long-term price trends, propelling cryptocurrencies into a new phase of development.

02 Feature comparison: Conclusion

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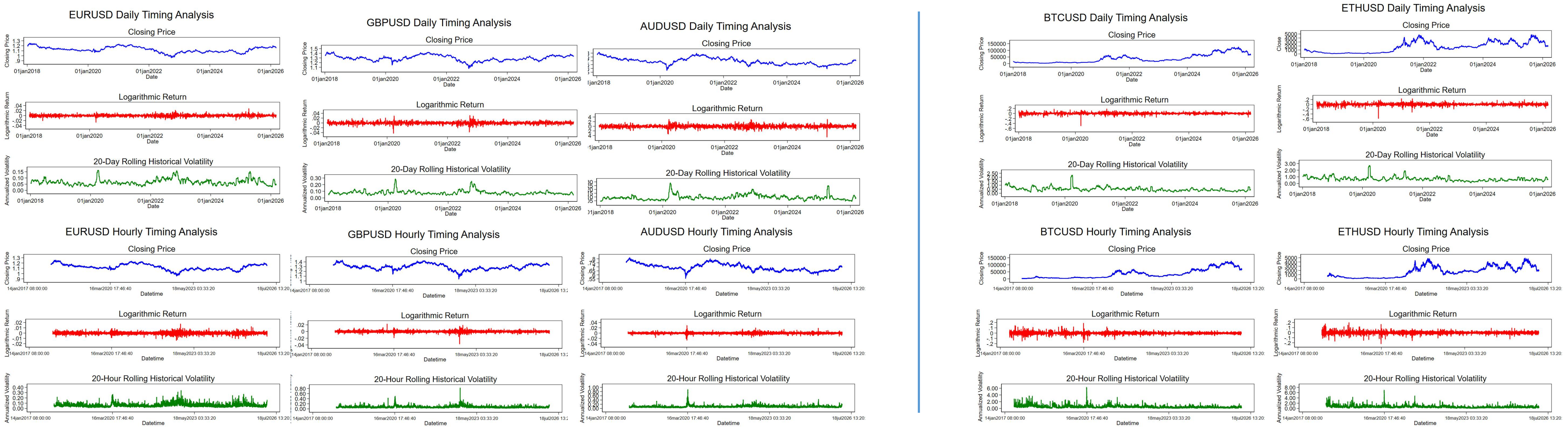
The characteristic of cryptocurrencies being **decentralized** makes it impossible to anchor, and their **overall volatility is greater than that of traditional currencies.**



dimensionality	Traditional currency（AUD/EUR/GBP）	Cryptocurrency（BTC/ETH）
Price trend	Long-term range-bound fluctuations with small amplitude (exchange rate between 0.6 and 1.8)	A long-term one-sided rise followed by a sharp pullback, with significant gains and losses (BTC rose from around 10,000 to over 130,000)
Yield volatility	The daily/hourly yield rates are all concentrated within ±0.04 (±4%), with very few extreme values	The daily yield rate can exceed ±0.6 (±60%), and the hourly yield rate can exceed ±0.2 (±20%), with extremely strong characteristics of sharp peaks and thick tails
Volatility level	The annualized volatility has remained stable at 0-0.3 (0-30%) over the long term, with a peak of approximately 0.3 during the crisis period	The annualized volatility has long been between 0.5 and 1.5 (50%-150%), and the hourly data peak can exceed 6.0 (600%).
Wave-driven	Driven by macroeconomics, central bank policies and geopolitics, the trend is slow and predictable	Driven by market sentiment, regulatory policies, and the crypto cycle, trends switch extremely rapidly

Time Series Testing

To ensure data stationarity and avoid spurious regression, so as to guarantee the reliability of machine learning forecasting results.

03 Time series testing (Part I): ADF（Augmented Dickey-Fuller） stationary test

Table 2: ADF Stationarity Test Results

Currency	Test statistic	Dickey-Fuller value	P-value	Lag order	Conclusion
BTC	Unit root test	-12.704	0.0000	10	Stable
ETH		-12.911	0.0000	10	Stable
AUD		-8.725	0.0000	3	Stable
EUR		-12.315	0.0000	3	Stable
GBP		-9.356	0.0000	3	Stable

The stationarity of the series was evaluated using the ADF (Augmented Dickey-Fuller) test, and the results show that all currencies **reject the null hypothesis** of 'presence of a unit root' at the 1% significance level. The log return series is stationary, meeting the prerequisites for GARCH model construction.

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01

The autocorrelation coefficients of **traditional currencies** and **Bitcoin** for all lags (from Lag 1 to 20) **almost all fall within the grey confidence interval**.Only a few points slightly exceed the interval, but the amplitude is extremely small.(next page)

indicates

there is **no significant linear autocorrelation in the sequence**, which is in line with the characteristic of **"return rates being unpredictable"** under the weak efficient market hypothesis.

02

Ethereum has many points slightly **exceeding** the interval.

indicates

there is **significant linear autocorrelation in the sequence**. The Q test of the Ljung-Box statistic also shows this result.

Table 3: The Result of the Ljung-Box Test for Autocorrelation

Currency	Lag order	Q statistic	p-value	Conclusion
BTC	20	20.479	0.4283	no autocorrelation
ETH	20	40.049	0.0049	autocorrelation
AUD	20	18.695	0.5417	no autocorrelation
EUR	20	15.652	0.7380	no autocorrelation
GBP	20	15.939	0.7204	no autocorrelation

03 Time series testing (Part II) : Autocorrelation Test (ACF+Ljung-Box)

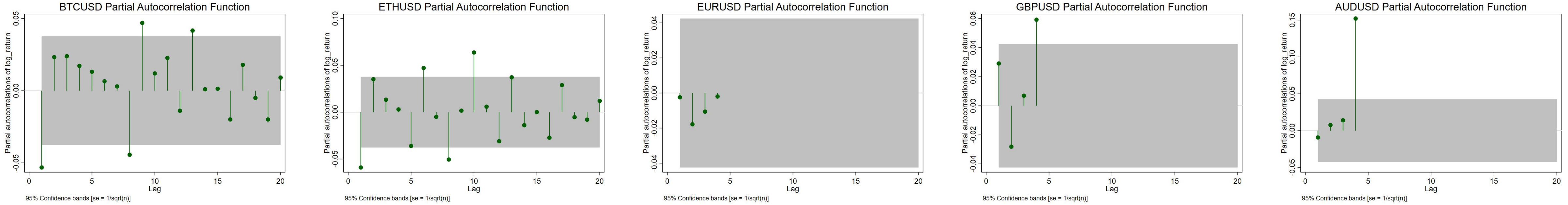

Figure 4：partial autocorrelation

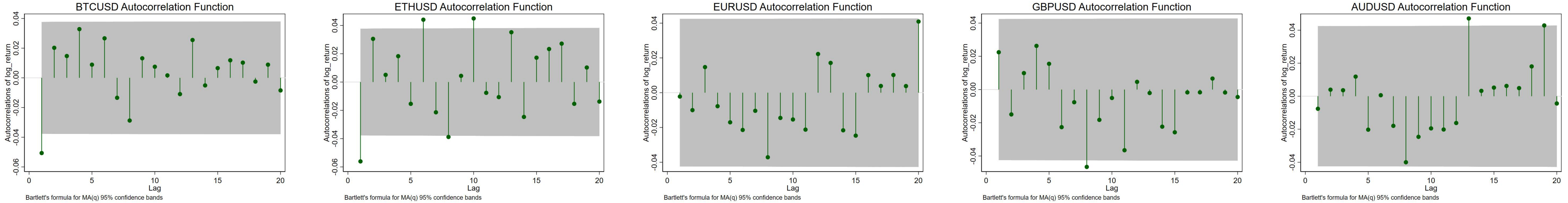

Figure 5：autocorrelation

Table 4: Logarithmic Return ARCH Effect Test

Currency	Chi-square	P-value
BTC	9.732	0.0018
ETH	22.244	0.0000
AUD	32.311	0.0000
EUR	17.406	0.0000
GBP	48.238	0.0000

Purpose of inspection

Verifying the presence of heteroscedasticity in the residual of the yield rate series is a critical prerequisite for volatility modeling using GARCH models.

Result of inspection

The LM test revealed that the residual series of all five assets rejected the null hypothesis at the 1% significance level, indicating the presence of a significant ARCH effect.

Key conclusion

The data characteristics fully satisfy the applicability criteria of the GARCH model, making this model family a reliable choice for capturing and forecasting volatility.

Currency Fluctuation Prediction

Using some machine learning models to **analyze and compare the prediction results** of the volatility of cryptocurrencies and traditional currencies based on low-frequency (daily) data.

Data Type：Daily Frequency Data

Time Span：2017-12-12 to 2026-03-09

Object of Study：Cryptocurrency(ETHUSD、BTCUSD);
Traditional currency(EURUSD、GBPUSD、AUDUSD)

Dataset division：Training set: Test set = 8:2

Evaluation metrics: Mean Absolute Error (MAE), Root mean square Error (RMSE)
- **The lower the error, the better the model's prediction effect**

Based on the above design, a single machine learning and GARCH hybrid model is constructed to conduct empirical research on volatility prediction

Table 5: Model Types

Category	Model	Description
Single Models	SVR	Support Vector Regression, captures nonlinear relationships via kernel methods, suitable for small-sample high-dimensional data
	XGBoost	Gradient boosting tree, iteratively optimizes residuals, incorporates regularization to prevent overfitting
	Random Forest	Ensemble learning model, aggregates multiple decision trees for prediction, robust to outliers
Hybrid Models	GARCH-SVR	Conditional volatility extracted by GARCH is used as a feature, then input into SVR for refined prediction
	GARCH-Random Forest	Conditional volatility extracted by GARCH is used as a feature, then input into Random Forest for refined prediction
	GARCH-XGBoost	Conditional volatility extracted by GARCH is used as a feature, then input into XGBoost for refined prediction

The hybrid model integrates GARCH's **temporal feature capture ability** with machine learning's **nonlinear fitting ability**.

04 Currency fluctuation prediction：Model prediction results

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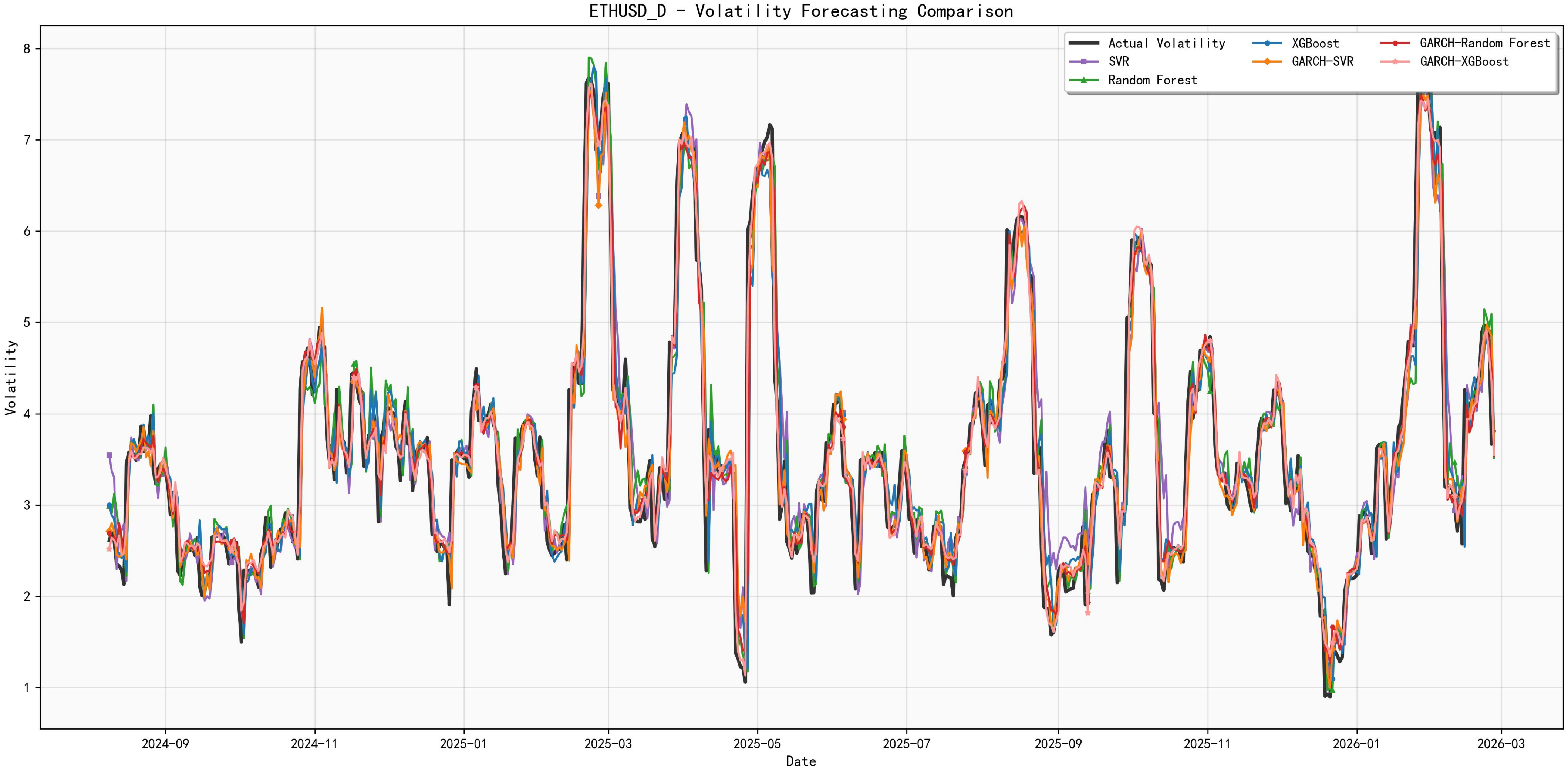

Figure 6：Model Fitting and Prediction Results
（ETHUSD_D_forecast_comparison）

04 Currency fluctuation prediction：Model prediction results

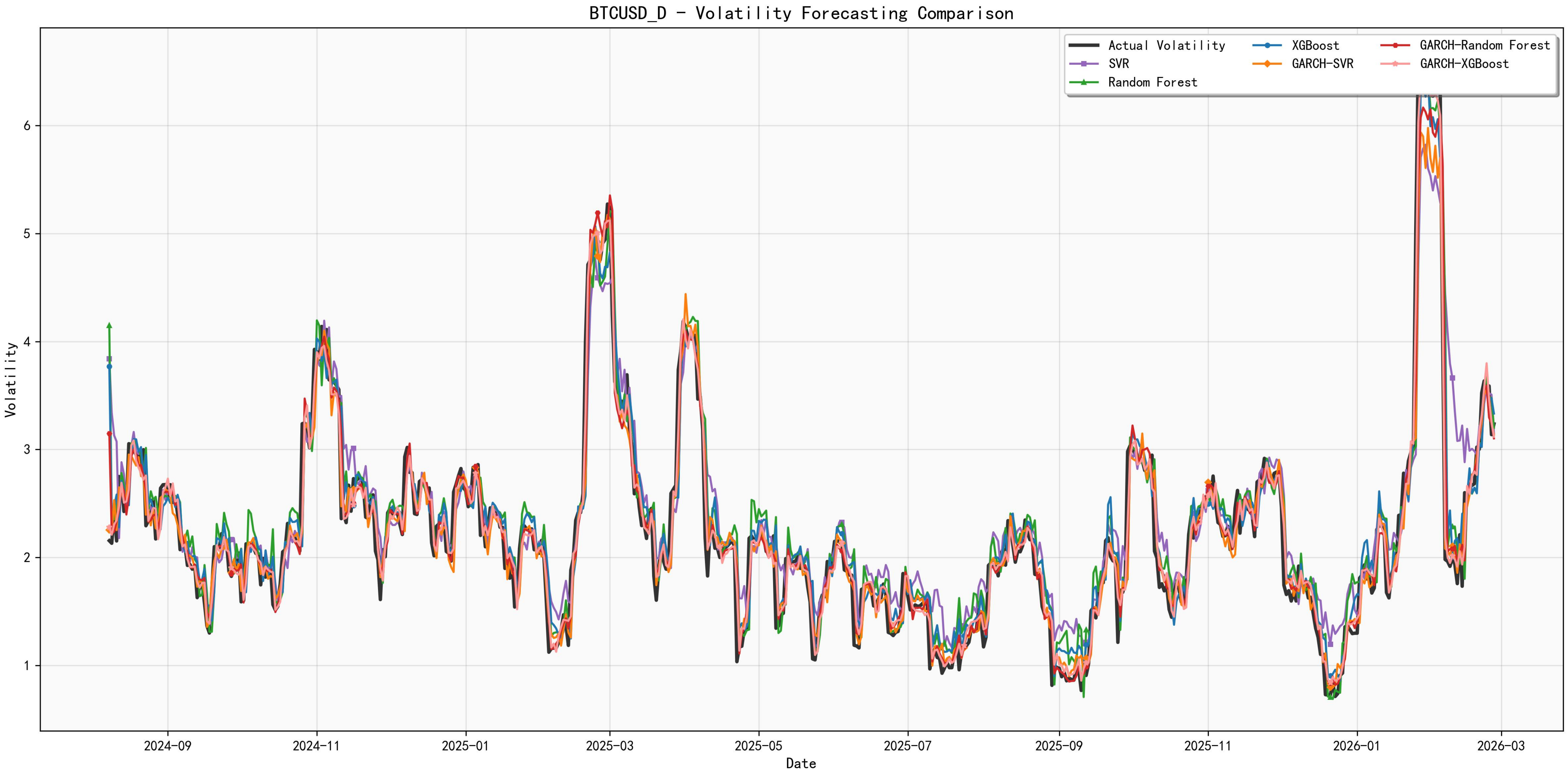

Figure 7：Model Fitting and Prediction Results
（BTCUSD_D_forecast_comparison）

04 Currency fluctuation prediction：Model prediction results

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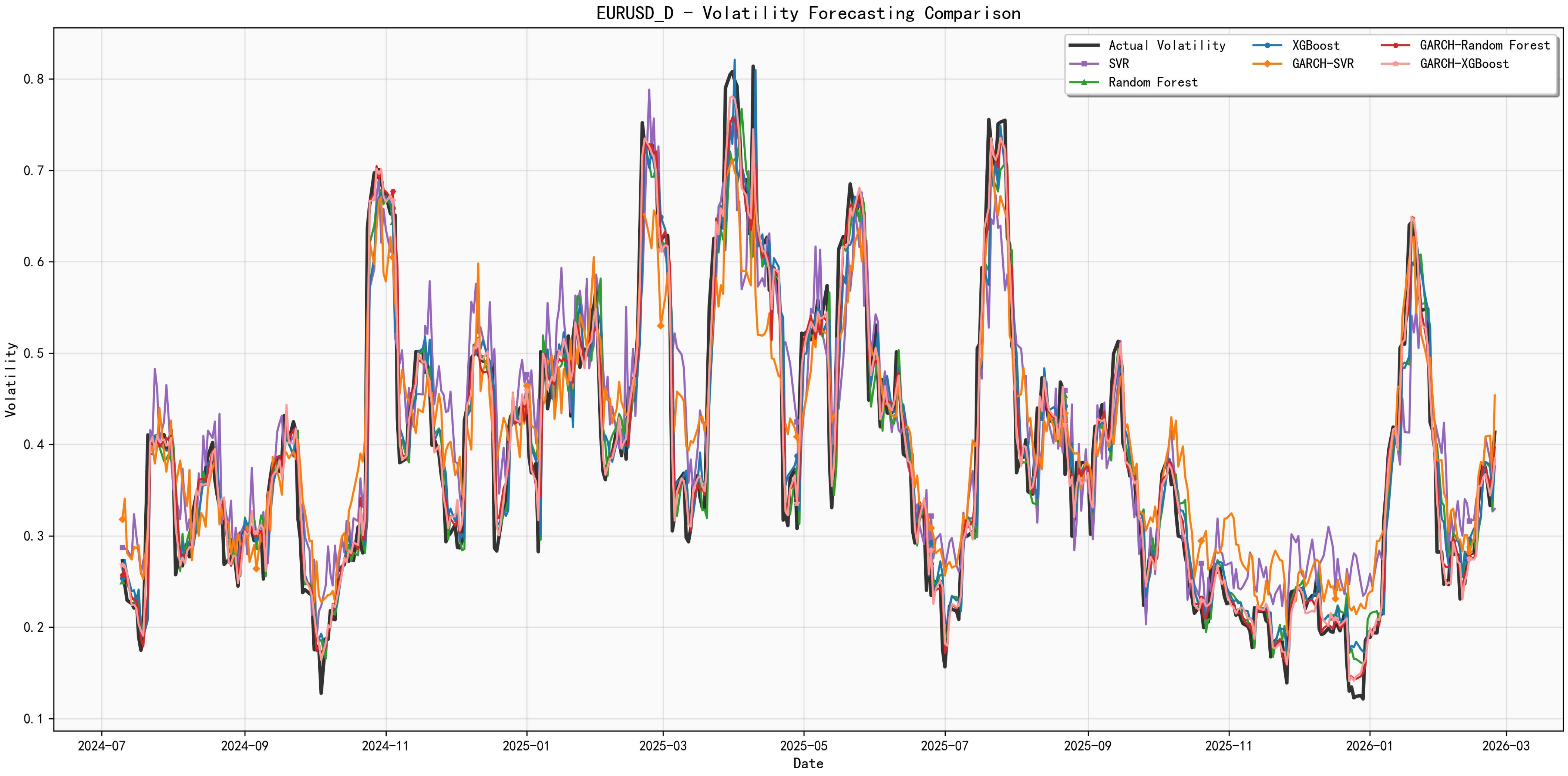

Figure 8：Model Fitting and Prediction Results
（EURUSD_D_forecast_comparison）

04 Currency fluctuation prediction：Model prediction results

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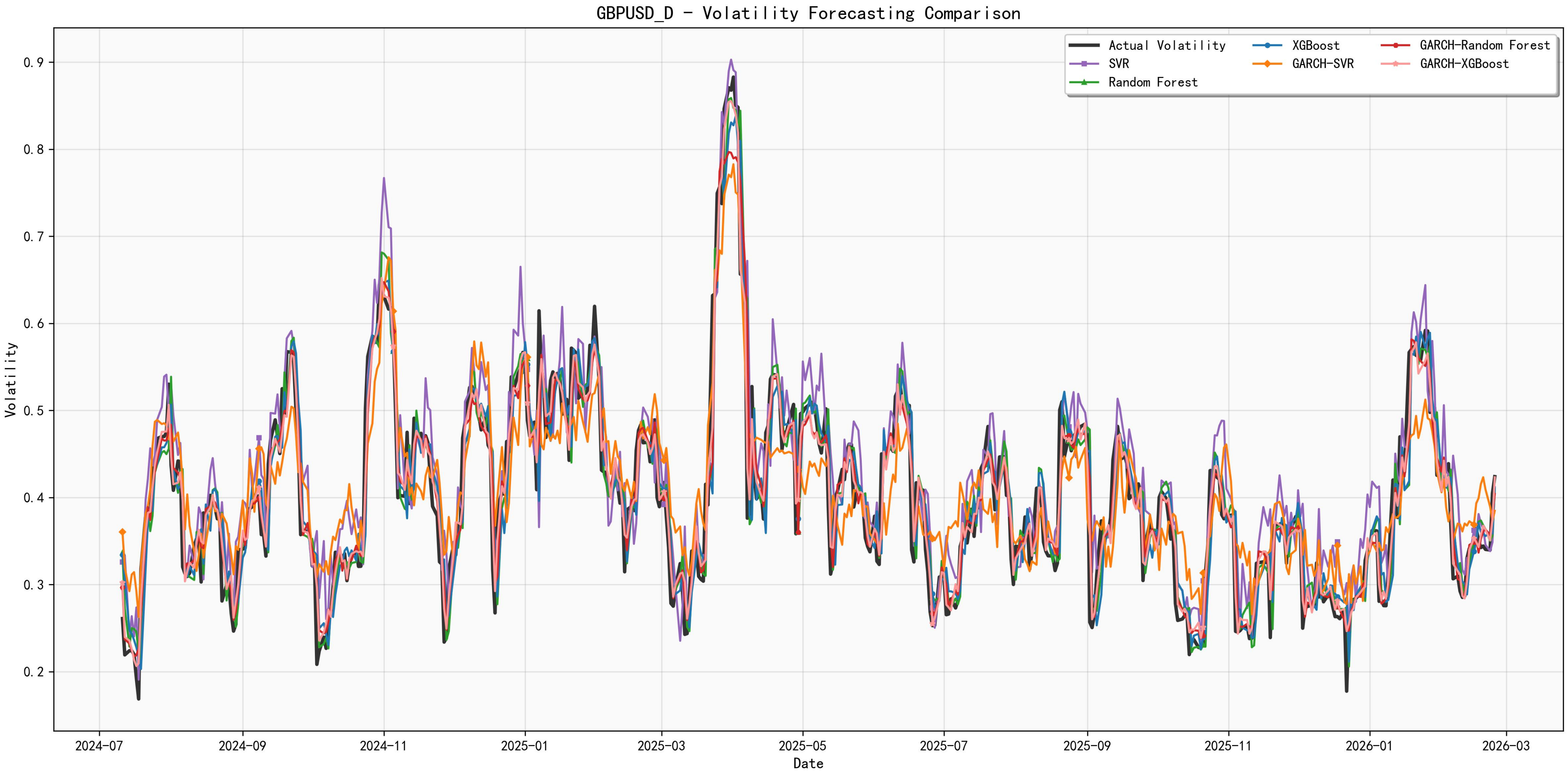

Figure 9：Model Fitting and Prediction Results
（GBPUSD_D_forecast_comparison）

04 Currency fluctuation prediction：Model prediction results

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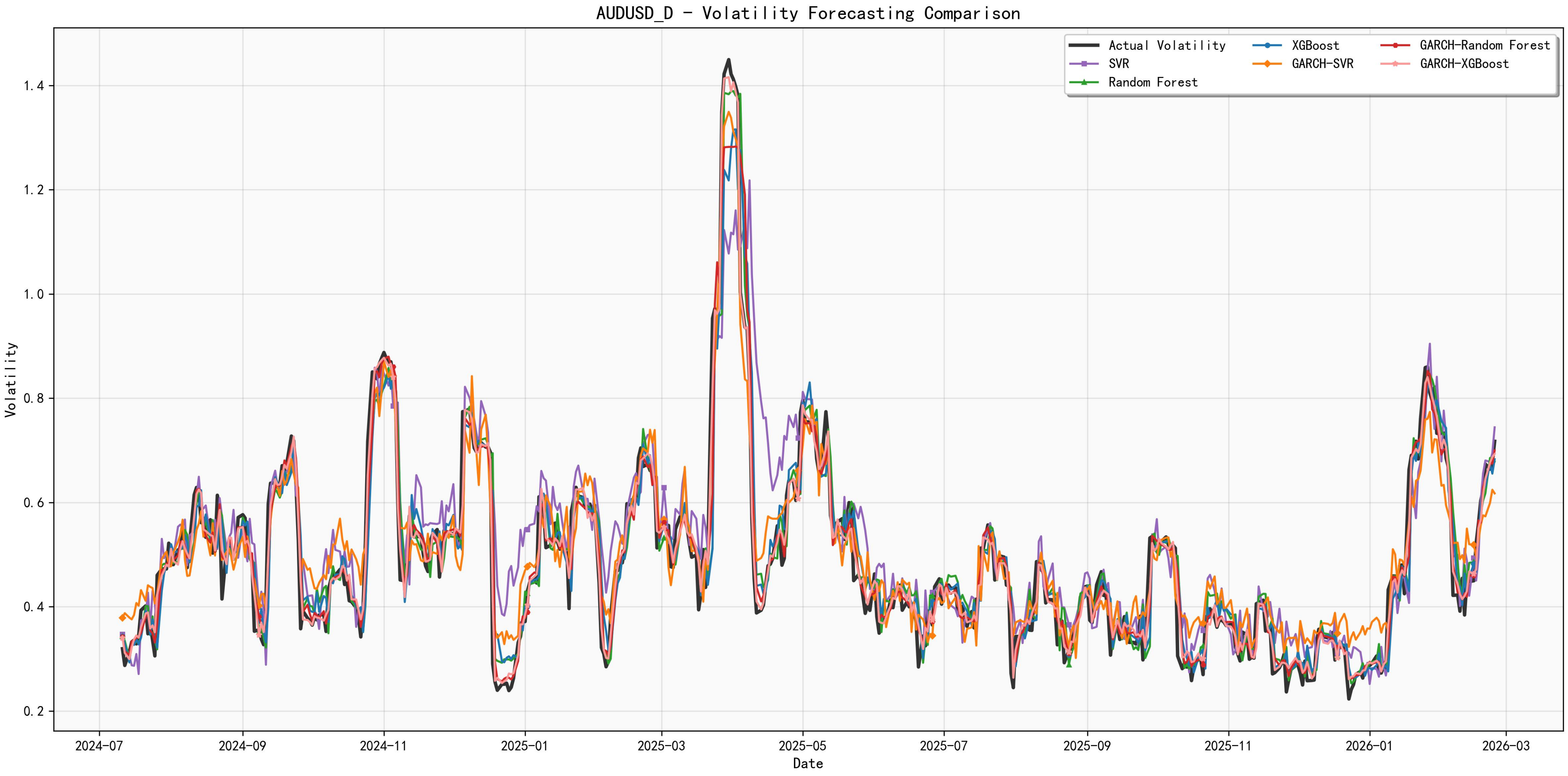

Figure 10： Model Fitting and Prediction Results
（AUDUSD_D_forecast_comparison）

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XGBoost demonstrates strong fitting capability

This result demonstrates that XGBoost, as a boosting ensemble method, possesses **stronger fitting capabilities** when handling complex non-linear relationships through iterative optimisation of residuals and the introduction of regularisation mechanisms, enabling it to make fuller use of the conditional heteroscedasticity information extracted by the GARCH model.

GARCH-XGBoost demonstrates superior performance

In comparison, **GARCH-Random Forest ranks second**, whilst GARCH-SVR performs relatively weakly among the hybrid models, though it still represents **a significant improvement over the standalone SVR model**. As can be seen from the fitting and forecasting results in Figure6-10, all hybrid models capture the overall trend of volatility well, with **GARCH-XGBoost providing a more precise fit to the finer details of volatility**, demonstrating particularly strong adaptability during periods of high volatility.

GARCH-XGBoost ＞ GARCH-Random Forest ＞ GARCH-SVR

Table 6: Forecasting Results of Different Models for Daily Data

	Cryptocurrency Group				Traditional Currency Group					
	ETHUSD		BTCUSD		EURUSD		GBPUSD		AUDUSD	
MODELS	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
SVR	**0.347**	**0.544**	**0.285**	**0.419**	0.057	0.073	0.045	0.059	0.067	0.100
Random Forest	**0.325**	**0.532**	**0.222**	**0.339**	0.032	0.050	0.032	0.048	0.039	0.065
XGBoost	**0.325**	**0.522**	**0.205**	**0.311**	0.033	0.050	0.032	0.048	0.042	0.064
GARCH-SVR	**0.244**	**0.408**	**0.148**	**0.241**	0.052	0.062	0.050	0.059	0.053	0.064
GARCH-Random Forest	**0.219**	**0.360**	**0.136**	**0.223**	0.021	0.035	0.022	0.034	0.026	0.046
GARCH-XGBoost	**0.190**	**0.274**	**0.124**	**0.178**	0.019	0.027	0.018	0.026	0.021	0.031

conclusion：

- **The prediction error of the cryptocurrency group is significantly higher** than that of the traditional currency group, which conforms to the **characteristics of high volatility**;

Table 7: Cryptocurrency Error Reduction Rate

model	MAE	RMSE	The decline rate compared with GARCH-XGBoost (MAE/RMSE)
SVR	0.347	0.544	45.24% / 49.63%
Random Forest	0.325	0.532	41.54% / 48.50%
XGBoost	0.325	0.522	41.54% / 47.51%
GARCH-SVR	0.244	0.408	- / -
GARCH-RF	0.219	0.360	- / -
GARCH-XGBoost	0.190	0.274	optimal

model error of ETHUSD

model	MAE	RMSE	The decline rate compared with GARCH-XGBoost (MAE/RMSE)
SVR	0.285	0.419	56.49% / 57.52%
Random Forest	0.222	0.339	44.14% / 47.49%
XGBoost	0.205	0.311	39.51% / 42.77%
GARCH-SVR	0.148	0.241	- / -
GARCH-RF	0.136	0.223	- / -
GARCH-XGBoost	0.124	0.178	optimal

model error of BTCUSD

conclusion:

- The hybrid method that combines the GARCH model with machine learning models can effectively **capture the temporal characteristics of volatility** and the **complex nonlinear patterns**, significantly improving the prediction accuracy.

04 Currency fluctuation prediction：Model prediction results

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Table 8: Traditional Currency Reduction Rate

model	MAE	RMSE	The decline rate compared with GARCH-XGBoost (MAE/RMSE)
SVR	0.057	0.073	67%/63%
Random Forest	0.032	0.050	41%/46%
XGBoost	0.033	0.050	42%/46%
GARCH-SVR	0.052	0.062	- / -
GARCH-RF	0.021	0.035	- / -
GARCH-XGBoost	0.019	0.027	optimal

model error of EURUSD

model	MAE	RMSE	The decline rate compared with GARCH-XGBoost (MAE/RMSE)
SVR	0.045	0.059	60%/ 55.93%
Random Forest	0.032	0.048	43.75%/ 45.83%
XGBoost	0.032	0.048	43.75%/ 45.83%
GARCH-SVR	0.050	0.059	- / -
GARCH-RF	0.022	0.034	- / -
GARCH-XGBoost	0.018	0.026	optimal

model error of GBPUSD

model	MAE	RMSE	The decline rate compared with GARCH-XGBoost (MAE/RMSE)
SVR	0.067	0.100	68.66% /69%
Random Forest	0.039	0.065	46.15%/ 52.31%
XGBoost	0.042	0.064	50%/ 51.56%
GARCH-SVR	0.053	0.064	- / -
GARCH-RF	0.026	0.046	- / -
GARCH-XGBoost	0.021	0.031	optimal

model error of AUDUSD

conclusion:

- The GARCH-XGBoost model demonstrates **significantly higher average prediction accuracy improvement** **for traditional currency** groups compared to its performance on cryptocurrency groups. This indicates that for traditional currencies with relatively stable volatility and clearer structural patterns, the feature fusion advantages of hybrid models can be more fully utilized.

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Table 9: Forecasting Results of Different Models for Daily Data

	Cryptocurrency Group				Traditional Currency Group					
	ETHUSD		BTCUSD		EURUSD		GBPUSD		AUDUSD	
MODELS	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
SVR	0.347	0.544	0.285	0.419	0.057	0.073	0.045	0.059	0.067	0.100
Random Forest	0.325	0.532	0.222	0.339	0.032	0.050	0.032	0.048	0.039	0.065
XGBoost	0.325	0.522	0.205	0.311	0.033	0.050	0.032	0.048	0.042	0.064
GARCH-SVR	0.244	0.408	0.148	0.241	0.052	0.062	0.050	0.059	0.053	0.064
GARCH-Random Forest	0.219	0.360	0.136	0.223	0.021	0.035	0.022	0.034	0.026	0.046
GARCH-XGBoost	**0.190**	**0.274**	**0.124**	**0.178**	**0.019**	**0.027**	**0.018**	**0.026**	**0.021**	**0.031**

conclusion：

- The **GARCH hybrid model** **outperforms** the corresponding **single machine learning model** in all currency pairs；
- **GARCH-XGBoost** is the **optimal model** among all models, with the **lowest MAE/RMSE.**

H1 is correct:

Model fusion validity assumption：Hybrid models (such as GARCH-XGBoost) demonstrate significantly superior prediction accuracy for volatility compared to standalone machine learning models..

- The out-of-sample prediction results roughly conform to the in-sample fitting situation, and each mixed model demonstrates good generalization ability.
- Overall, the GARCH-XGBoost model has significant advantages in volatility prediction, capable of more accurately capturing the dynamic change characteristics of volatility and providing an effective modeling method for the volatility prediction of the money market.
- The above results further verify the feasibility and superiority of combining traditional time series models with machine learning methods.

H2 is correct:

Hypothesis on the Impact of Asset Heterogeneity：The hybrid model demonstrates varying performance in volatility prediction for two asset classes, with differing improvement magnitudes observed.

Correlation Analysis

By combining the Dynamic Conditional Correlation (DCC) and Dynamic Correlation and Network Topology Feature Analysis, the correlation of volatility between cryptocurrencies and traditional currencies is characterized and analyzed.

05 Correlation analysis：DCC GARCH model

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Study subjects: Cryptocurrencies (BTC/ETH) vs Traditional currencies (AUD/EUR/GBP)

Research method：DCC-GARCH(1,1) **dynamic conditionally correlated model**

Research emphasis：

- Time-varying Volatility Correlation Characteristics,
- Structural Changes Before and After the Pandemic
- Volatility Persistence and Model Testing

Core Objective: Analysis of Cross-Market Volatility Spillovers and Linkages

Model Robustness Testing:

- Univariate GARCH: Both ARCH and GARCH terms are statistically significant, confirming the presence of volatility clustering.
- DCC component: satisfies the stationarity condition and exhibits stable dynamic correlation structure
- Distribution Selection: t-Distribution Effectively Captures Thick-Tail Characteristics of Financial Returns
- Conclusion: The model is well-designed and the results are reliable.

Descriptive Statistics

- Cryptocurrencies exhibit high correlation (0.85)
- Cryptocurrencies generally exhibit **weak** to moderate **positive** correlations with fiat currencies, fluctuating within the range of **0 to 0.4**.

Table 10：Descriptive statistics of currency correlation

	Coef.	Std. Err.	z	P>|z|	[95% Conf.	Interval]
corr(ret_ETH, ret_BTC)	.8532023	.0149142	57.21	0.000	.823971	.8824336
corr(ret_ETH, ret_AUD)	.258118	.0486522	5.31	0.000	.1627614	.3534746
corr(ret_ETH, ret_EUR)	.1828587	.0504882	3.62	0.000	.0839037	.2818138
corr(ret_ETH, ret_GBP)	.1943797	.050405	3.86	0.000	.0955876	.2931717
corr(ret_BTC, ret_AUD)	.2565703	.0494603	5.19	0.000	.1596299	.3535107
corr(ret_BTC, ret_EUR)	.1804137	.0508175	3.55	0.000	.0808133	.2800142
corr(ret_BTC, ret_GBP)	.1932694	.050764	3.81	0.000	.0937738	.292765
corr(ret_AUD, ret_EUR)	.6613351	.0287102	23.03	0.000	.6050642	.7176061
corr(ret_AUD, ret_GBP)	.6605851	.0286121	23.09	0.000	.6045064	.7166637
corr(ret_EUR, ret_GBP)	.7376016	.023627	31.22	0.000	.6912936	.7839096

05 Correlation analysis：Empirical Results

Ethereum (ETH) and Bitcoin (BTC) show highly synchronized correlation sequences with their respective fiat counterparts, indicating similar price movements between major cryptocurrencies and foreign exchange markets.

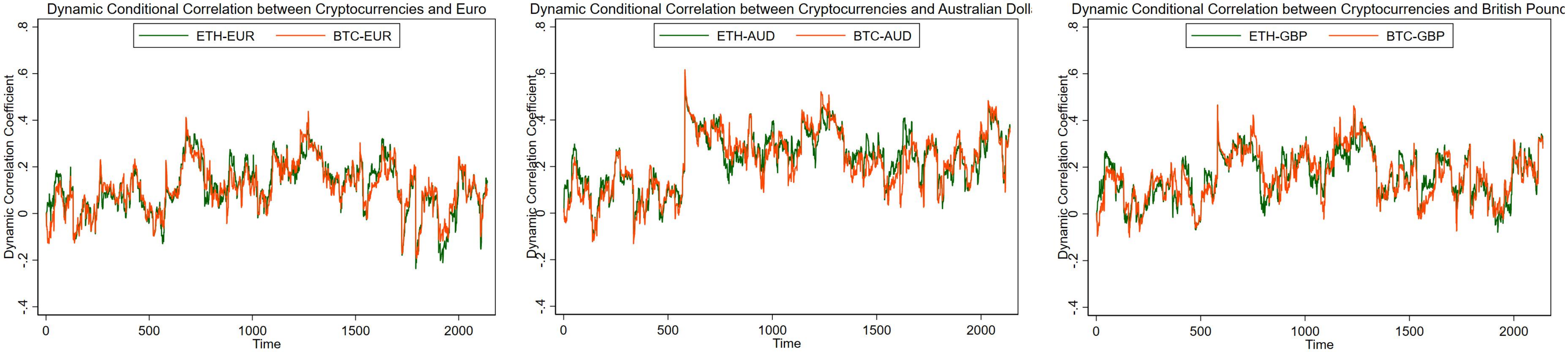

Figure 11： Dynamic Conditional Correlation between Cryptocurrencies and Traditionalcurrencies(a)

Key characteristics:

- **Time-varying nature**: Continuously changes over time;
- **Aggregation**: High correlation/low correlation phases appear in clusters;
- **Synchronization**: BTC and ETH exhibit highly consistent correlation trends with the same currency.

Specific time points:

- Correlative peaks emerge during the early stages of the pandemic;
- Cross-market linkages significantly strengthen when global uncertainties increase.

05 Correlation analysis：Pandemic change correlation

Partition nodes：**11 March 2020** (WHO declares global pandemic)
Sample segmentation：

- Pre-pandemic
- Post-pandemic

Phenomenon Description(**Box plot**)：

- Both the mean and median levels of correlations are **higher in the post-pandemic period**, accompanied by wider interquartile ranges and more extreme observations.
- Among the three fiat currencies, **AUD shows the most pronounced rise in correlation with both ETH and BTC**, while EUR and GBP display similar but relatively moderate upward shifts.

Conclusion：

- All **relevant indicators** showed **significant increases** post-pandemic.
- The pandemic has exacerbated the volatility **spillover effects** between cryptocurrencies and foreign exchange markets.

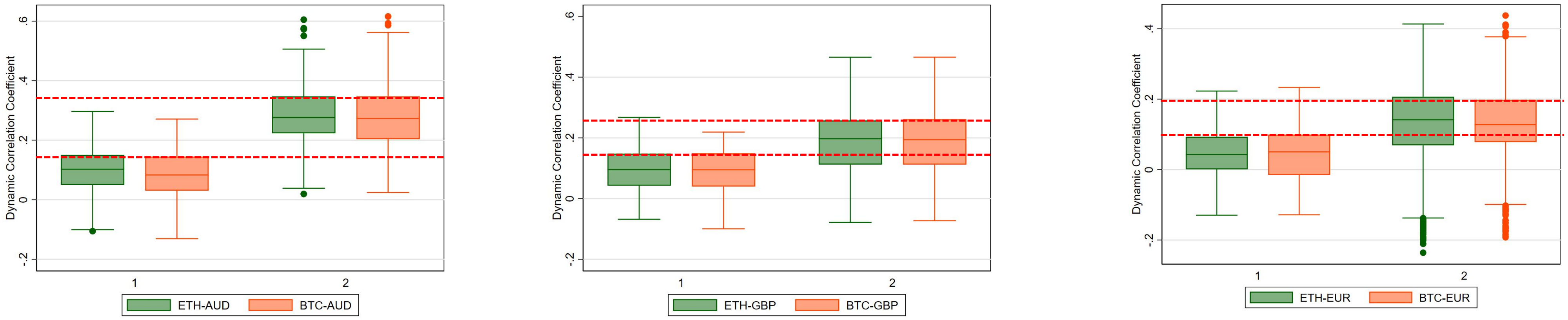

Figure 12：Dynamic Conditional Correlation between Cryptocurrencies and Traditionalcurrencies(b)

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Level of correlation

The overall correlation exhibited weak to **moderate positive strength**, with coefficient fluctuations concentrated within the range of 0 to 0.4.

Key features(H3)

It exhibits pronounced **time-varying characteristics and clustering behavior**; BTC and ETH show highly synchronized correlation trends with traditional currencies.

Performance at specific time points

During periods of heightened global uncertainty (such as the early stages of the pandemic), cross-market risk linkages significantly intensified, resulting in spikes in correlation coefficients.

Economic Implication: In the post-pandemic era, the increased co-volatility between cryptocurrencies and traditional monetary assets indicates **a decline in their investment diversification benefits**.

05 Correlation analysis：Dynamic Correlation and Network Topology Feature Analysis

- **Cryptocurrencies** exhibit consistently **high internal correlations**: The BTCUSD-ETHUSD correlation chart shows a persistent dark red zone with stable coefficients ranging from 0.8 to 0.9, demonstrating strong synchronized price movements within the crypto market and confirming their **highly correlated** asset portfolio nature.
- **Traditional currency** pairs demonstrate **stable internal correlations**: EURUSD, GBPUSD, and AUDUSD have maintained **medium-to-high** positive correlations over the long term (orange-red coloration), consistent with the linkage patterns observed among developed economies' currencies. **Post-pandemic**, these correlations have shown **a slight increase**.
- The correlation between fiat currencies and cryptocurrencies exhibited **significant volatility**: cross-market pairs such as EURUSD/BTCUSD and GBPUSD/BTCUSD displayed dramatic color changes over time, with positive correlations markedly strengthening post-pandemic. This indicates a **substantial increase** in interconnectivity between these two asset classes **following the crisis**.

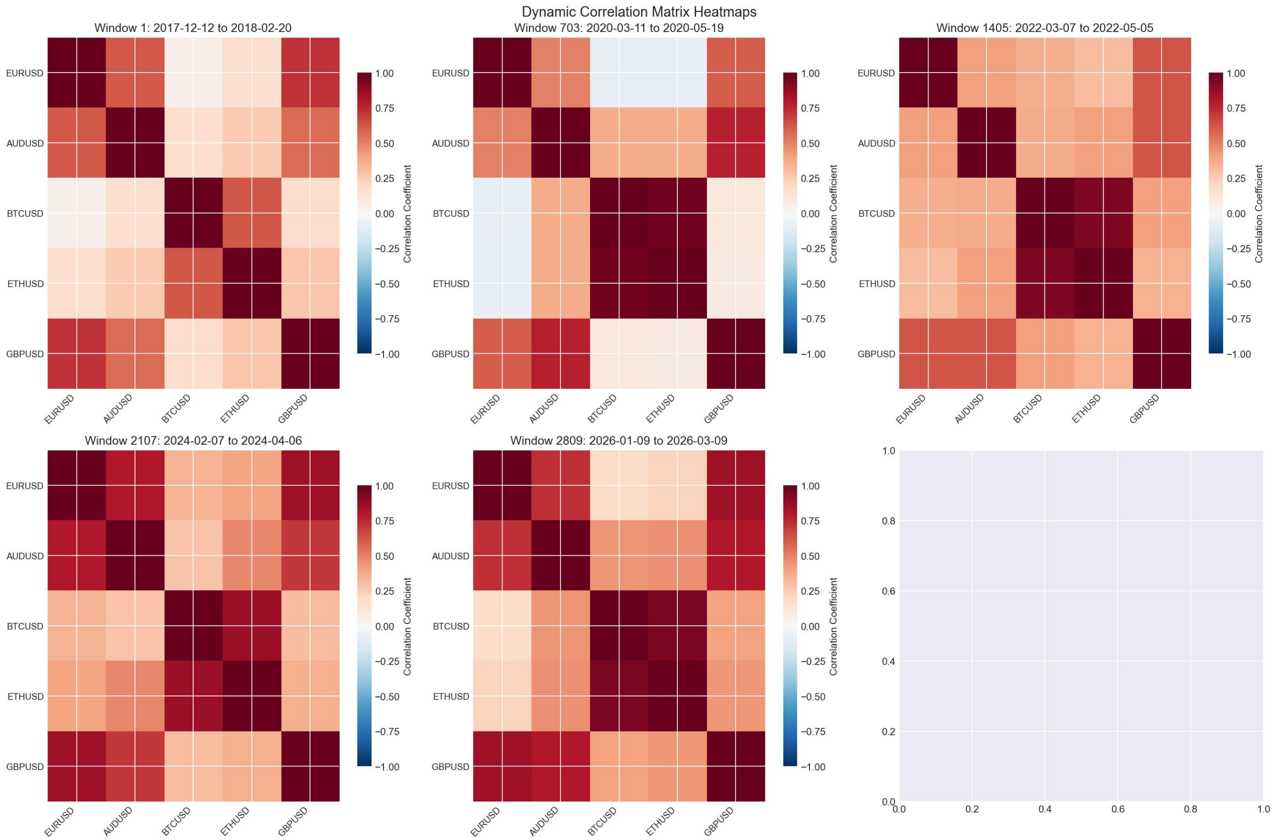

Figure 13：correlation_heatmaps

- Network interconnectedness shows **significant increase during crisis periods**: The simultaneous surge in **edge counts** and **network density** during the 2020 pandemic and 2022 Federal Reserve rate hikes demonstrates that extreme events substantially amplify **market-wide interconnectedness**.
- The **clustering coefficient** has remained persistently **high**: The **average clustering coefficient** stays stable between 0.5 and 0.9, indicating that assets consistently exhibit "**clustered linkage**" characteristics, with extreme peaks occurring during crises.
- **Risk contagion** accelerates during crises: **The average path length** peaks in 2020 and 2022, reflecting significantly enhanced **risk transmission efficiency** among assets during financial crises.
- **Cryptocurrencies** continue to demonstrate growing **systemic importance**: their **maximum centralization index** has repeatedly surpassed 0.9 since 2020, indicating that BTC/ETH are progressively becoming the network's core, with influence surpassing traditional currencies.
- The **Network Community Structure** has undergone temporal **differentiation**: **modularity** exhibited an overall downward trend after 2020, indicating that the boundaries between the 'fiat currency' and 'cryptocurrency' subgroups have become increasingly **blurred,** with continuous **enhancement** in the convergence degree of these two asset categories.

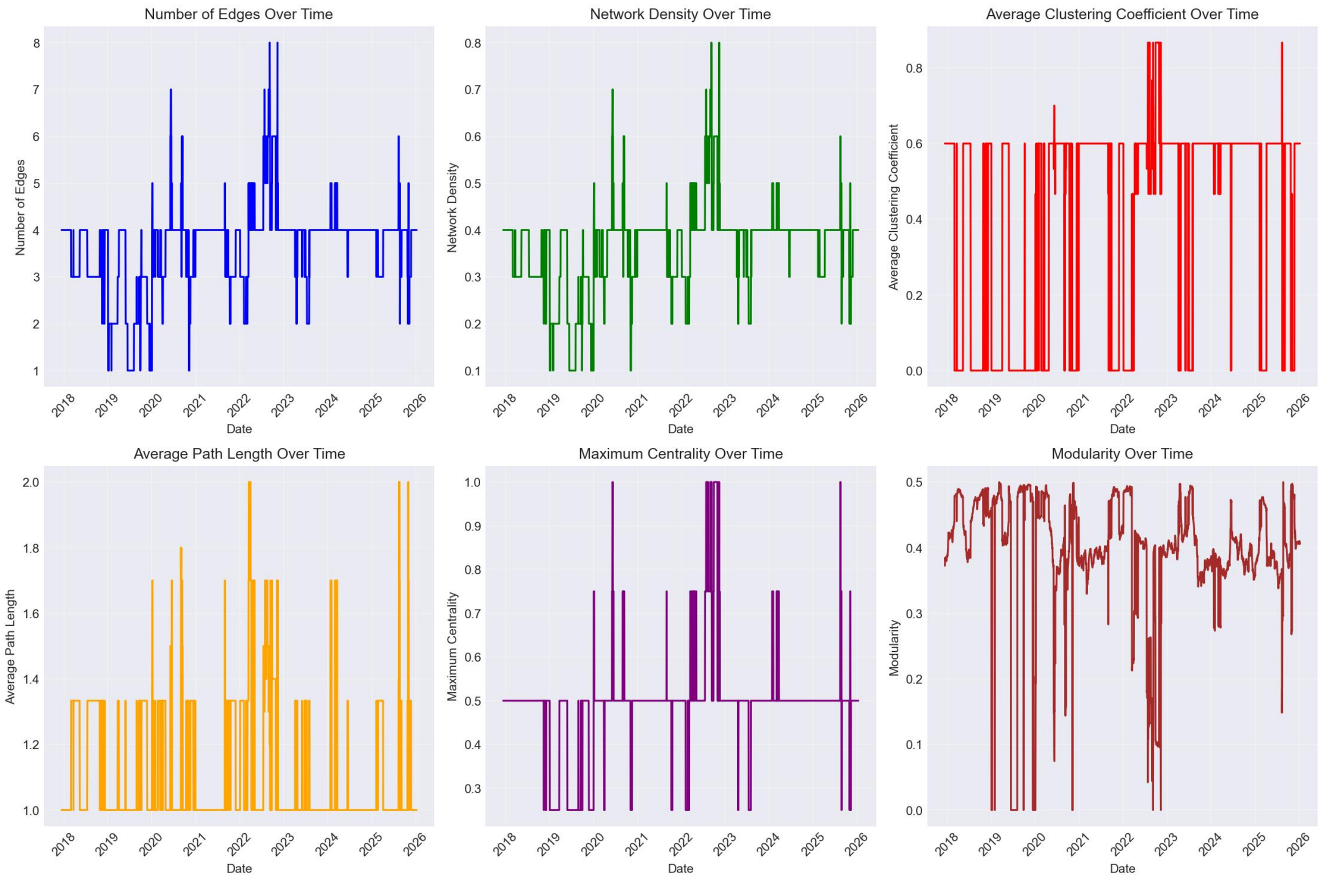

Figure 14：metrics_evolution

- Traditional currency subnetworks exhibit **long-term stability**: EURUSD, GBPUSD, and AUDUSD maintain strong connectivity (with persistent edges), forming a stable 'fiat currency core triangle.
- Cryptocurrency subnetworks have always maintained **independence**: The strong correlations between BTCUSD and ETHUSD remain consistently stable across all time windows, forming independent 'cryptocurrency core pair subnetworks' that never establish strong connections with fiat networks (correlation coefficients consistently <0.5).
- There is **no strong cross-market correlation** between fiat currencies and cryptocurrencies: In all analysis windows, no connections were observed between fiat currency nodes and crypto nodes, indicating that the correlation between these two asset classes has consistently remained below 0.5 over the long term, demonstrating **no significant linkage and maintaining risk isolation**.
- The network structure exhibited **localized adjustments** during extreme periods: In the 2022 window (upper right), the connection between AUDUSD and core fiat currencies weakened, reflecting the Australian dollar's reduced correlation with European currencies due to commodity price shocks during that period.

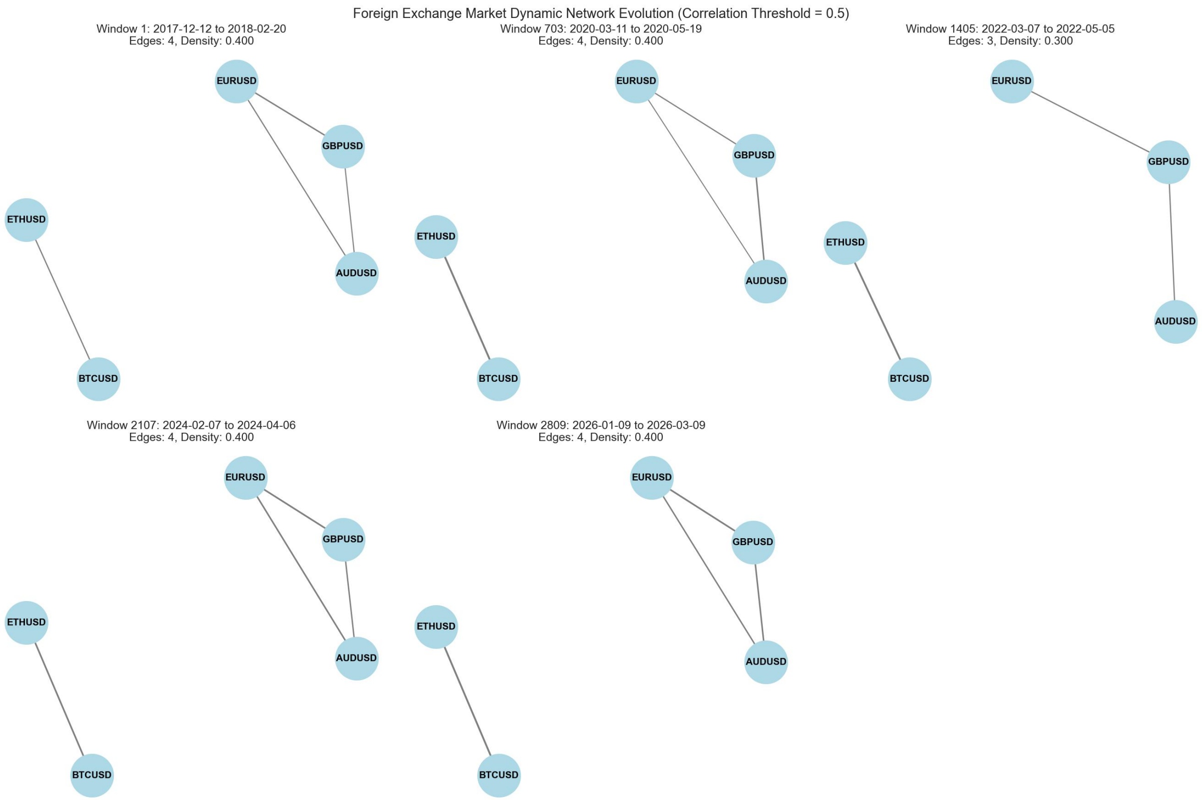

Figure 15：network_evolution

05 Correlation analysis: Dynamic Correlation and Network Topology Feature Analysis

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- **The green line** (BTCUSD-ETHUSD) demonstrates cryptocurrency intramarket correlation, maintaining **long-term stability at 0.8-0.9**—the highest correlation rate across all markets.
- **The blue line** (EURUSD-GBPUSD), **orange line** (EURUSD-AUDUSD), and **purple line** (GBPUSD-AUDUSD) illustrate traditional currency intramarket correlation, consistently ranging between 0.6-0.8 with **minimal volatility**.
- **The red line** (EURUSD-BTCUSD) represents fiat-crypto cross-market correlation, exhibiting **significant fluctuations within-0.2 to 0.6** over extended periods, marking the least stable linkage among all correlations.

- Extreme events triggered correlation mutations: During the pandemic collapse in **March 2020** and the Federal Reserve's interest rate hikes **in 2022**, all correlation coefficients experienced **significant fluctuations** simultaneously, reflecting how **market sentiment** dominated asset pricing across the entire market during crises.

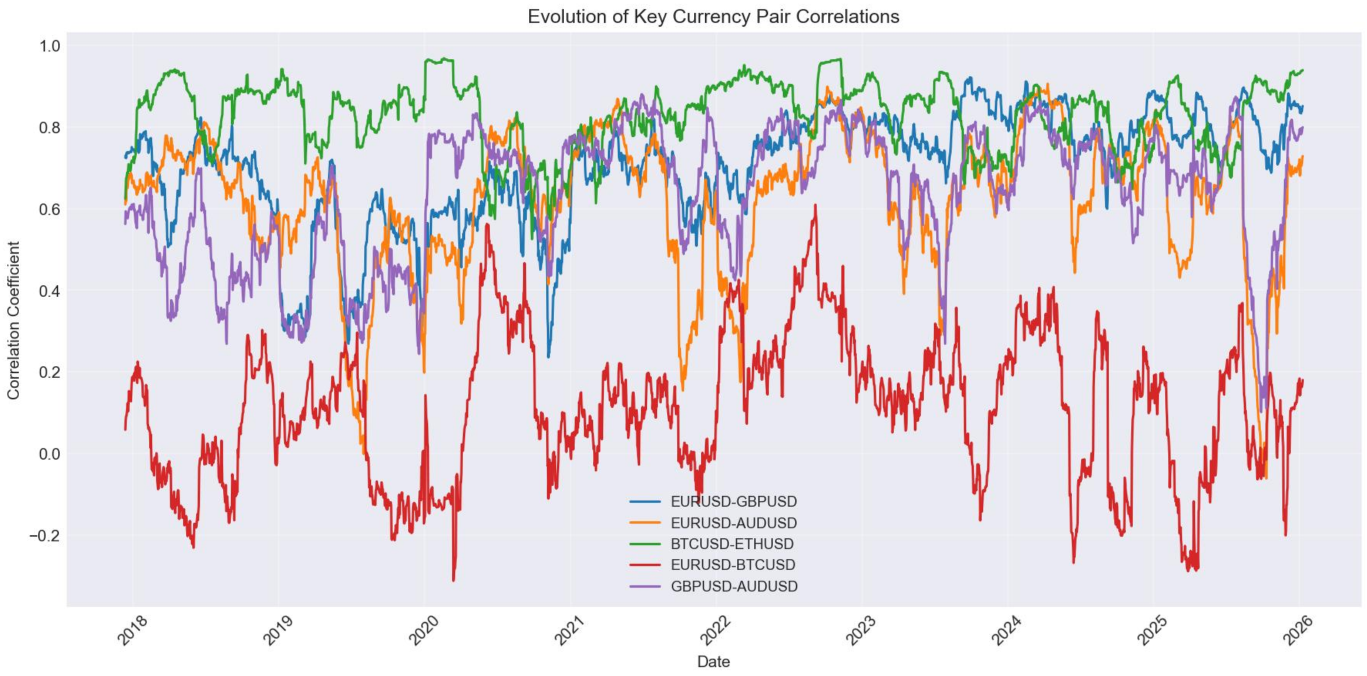

Figure 16: pairwise_correlation

Market structure stratification stability

Traditional currencies form a stable core sub-network, while cryptocurrencies constitute an independent yet strongly interconnected sub-network. These two asset classes maintain long-term risk isolation, with cross-market linkages being weaker than intra-market correlations.

Extreme events reshape connectivity

Exogenous shocks such as pandemics and interest rate hikes will significantly enhance market-wide interconnectedness, accelerate risk contagion, and strengthen cross-market linkages between fiat currencies and cryptocurrencies.

long-term trend

The integration level continues to rise, with the market boundaries between traditional currencies and cryptocurrencies becoming increasingly blurred, and their convergence steadily intensifying.

Research Conclusion

The advantages of the mixed model are significant(H1)

The GARCH-XGBoost hybrid model demonstrates superior performance in volatility prediction, with accuracy significantly outperforming standalone machine learning models.

Dynamic Correlation and Time-Varying Characteristics(H3)

Cryptocurrencies exhibit time-varying positive correlations with traditional currencies. Post-pandemic, their interdependence has significantly intensified, resulting in heightened cross-market risks.

Asset type influences the magnitude of improvement(H2)

The hybrid model demonstrates significantly greater improvement in prediction accuracy for traditional currency groups compared to cryptocurrency groups.

Significant differences in prediction difficulty

Cryptocurrencies pose greater challenges for prediction due to their high volatility and complexity.

Thanks for your watching!